

POLITECNICO
MILANO 1863

Visual reconstruction of played music

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I. Method

A. Overall Pipeline

B. Frame Selection and Preprocessing

C. Piano Key Detection

D. Fingertip Detection

E. Fingertip-to-Key Association

F. Pressed-Key Detection

II. Implementation

A. Frame Extraction and ROI Localization

B. White and Black Key Localization

C. Fingertip Detection

D. Key Mapping and Pressed-Key Logic

III. Experiments and Results

A. Dataset

B. Qualitative Results

C. Quantitative Evaluation

D. Failure Cases and Limitations